

AI-LABS — Artificial Intelligence (CS-323) — All Labs+ Viva

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Lab weight** ~20% · 13 labs · viva 5+10 Qs

Source workbook: Practical Workbook CS-323 (NED UET, Dept. of Computer & Information Systems Engineering, revised Aug 2024). **Credits:** 3+1 (3 lectures + 1 lab/week) · **CLO:** "Demonstrate the use of modern tools and techniques for developing intelligent systems. (C3, PLO-3)" **Exclusions:** Lab 14 (Complex Engineering Activity) is **excluded** per request. **Ordering note:** the workbook runs **ANN-first** (W1–5) ahead of the NN theory timeline (W10–11). Labs are self-contained (perceptron → backprop) and do not depend on the theory. GA has **no lab** — theory pillar only. W8 midterm and W15 final have no lab; the lab viva is a separate W15 slot.

Lab → week map

Real Week	Lab(s)	Workbook Lab	Topic Group
W1	1	Implementing Simple Neural Network Using Perceptron	ANN / ML (Labs 1–5)
W2	2	Developing an Artificial Neural Network (ANN) Using Perceptron	ANN / ML (Labs 1–5)
W3	3	Applying Data Preprocessing for ANN	ANN / ML (Labs 1–5)
W4	4	Developing ANN Using ADALINE	ANN / ML (Labs 1–5)
W5	5	Developing ANN Using Backward Propagation	ANN / ML (Labs 1–5)
W6	6	Applying Uninformed Searching Techniques	Searching (Labs 6–7)
W7	7	Applying Informed Searching Techniques	Searching (Labs 6–7)
W8	—	MIDTERM — no lab	—
W9	8	Solving CSPs by Enforcing Arc Consistency	CSP (Labs 8–9)
W10	9	Solving CSPs Using Stochastic Local Search	CSP (Labs 8–9)
W11	10	Developing Knowledge-Based Systems	Expert Systems (Labs 10–12)
W12	11	Constructing Complex Rule-Based Systems	Expert Systems (Labs 10–12)
W13	12	Practicing Data-Driven Programming in Expert Systems	Expert Systems (Labs 10–12)
W14	13	Developing Fuzzy Logic Based System	Fuzzy Logic (Lab 13)
W15	—	FINAL — no lab (lab viva separate slot)	—

Notes

- ANN-before-theory tension:** ANN labs (W1–5) run before NN theory (narrative W10–11). Official workbook ordering — the lab is self-contained and does not depend on the theory timeline.
- GA has no lab counterpart:** the workbook contains no GA lab; GA stays a **theory pillar** (narrative W6) — self-study with `aimacode/aima-python` + Gate Smashers.
- No GA/PSO/ACO lab:** evolutionary-computing content is absent from this workbook; the ANN block covers the ML requirement.
- Group breakdowns: `AI/03-Lab-Breakdowns/01-ANN-and-ML.md` ... `05-Fuzzy-Logic.md`. Per-lab GitHub repos: `AI/Lab-Resources.md`.

Exam relevance (lab ↔ theory bridge)

Lab(s)	Examinable?	Bridge
1–5 ANN / ML	✓ theory-linked	NN + backprop pillars (W10–11) → <code>Formula-Book.md</code> (perceptron, backprop) · <code>Fear-Killer-Packs.md</code> (AI-W11)

6–7 Searching	✓ theory-linked	A pillar (W3) → Formula-Book.md (A) · Fear-Killer-Packs.md (AI-W3)
8–9 CSP	✓ theory-linked	AC-3 / CSP formulation (W4) → Fear-Killer-Packs.md (AI-W4)
10–12 Expert Systems	✓ theory-linked	forward/backward chaining, conflict resolution, RETE (W9) → Fear-Killer-Packs.md (AI-W9)
13 Fuzzy	✓ theory-linked	Mamdani vs Sugeno, defuzzification (W12) → Fear-Killer-Packs.md (AI-W12)
GA (no lab)	⚠ theory only	no workbook lab — self-study aimacode/aima-python + Gate Smashers

Lab Group: ANN / ML (Labs 1–5)

Source: Practical Workbook CS-323 (NED UET, revised Aug 2024). CLO: "Demonstrate the use of modern tools and techniques for developing intelligent systems. (C3, PLO-3)"

Lab 1 — Implementing Simple Neural Network Using Perceptron

Goal: Build a single-layer perceptron; train on logic functions. **Steps:** Model neuron (weighted sum → activation); train AND/OR/NOT/NAND/NOR gates with step activation; observe linear separability. **Deliverable:** Working perceptron + accuracy table per gate. **Resources:** AI/Lab-Resources.md → Lab 1

Lab 2 — Developing an Artificial Neural Network (ANN) Using Perceptron

Goal: Generalize Lab 1 into a reusable ANN using the perceptron learning rule. **Steps:** Learning rate, weight update rule ($w \leftarrow w + \eta \cdot (t - y) \cdot x$); fit/predict API; test on a real dataset (e.g., breast cancer). **Deliverable:** ANN class from scratch + evaluation metrics. **Resources:** AI/Lab-Resources.md → Lab 2

Lab 3 — Applying Data Preprocessing for ANN

Goal: Preprocess data before training (this is what separates a working model from a broken one). **Steps:** Handle missing values (imputation); feature scaling (MinMax/normalization); one-hot encoding of categoricals; train/test split. **Deliverable:** Preprocessing pipeline applied to a raw dataset + before/after training comparison. **Resources:** AI/Lab-Resources.md → Lab 3

Lab 4 — Developing ANN Using ADALINE

Goal: Implement ADALINE (Widrow-Hoff / delta rule) with linear activation. **Steps:** Batch (GD) vs stochastic (SGD) variants; loss = SSE; compare with perceptron on same problem; predict X/O characters. **Deliverable:** ADALINE implementation + perceptron-vs-ADALINE comparison. **Resources:** AI/Lab-Resources.md → Lab 4

Lab 5 — Developing ANN Using Backward Propagation

Goal: Implement multi-layer feed-forward network with backpropagation. **Steps:** Forward pass; MSE loss; backward pass (chain rule); gradient descent weight updates; train on XOR / seeds dataset. **Deliverable:** Backprop network from scratch + training loss curve + accuracy. **Resources:** AI/Lab-Resources.md → Lab 5

Exam Relevance (ANN / ML)

- Perceptron XOR non-linearity, ADALINE delta rule, backprop chain rule = exam-relevant theory (pillar). See Formula-Book.md, Fear-Killer-Packs.md (AI-W11).

Lab Group: Searching (Labs 6–7)

Source: Practical Workbook CS-323 (NED UET, revised Aug 2024). CLO: "Demonstrate the use of modern tools and techniques

Lab 6 — Applying Uninformed Searching Techniques for Problem Solving

Goal: Solve problems with uninformed search. **Steps:** Implement BFS, DFS, UCS, depth-limited, iterative deepening; compare completeness/optimality; solve a classic toy problem (e.g., missionaries-and-cannibals, 8-puzzle). **Deliverable:** Uninformed search implementations + comparison table (time, space, completeness, optimality). **Resources:** AI/Lab-Resources.md → Lab 6

Lab 7 — Applying Informed Searching Techniques for Problem Solving

Goal: Solve problems with informed search. **Steps:** Implement greedy best-first and A; *choose heuristics (Manhattan, Euclidean); verify admissibility; solve 8-/15-puzzle.* **Deliverable:** A implementation + heuristic comparison table (nodes expanded, solution cost). **Resources:** AI/Lab-Resources.md → Lab 7

Exam Relevance (Searching)

- A* = exam pillar (f = g + h, admissibility, optimality). See Formula-Book.md, Fear-Killer-Packs.md (AI-W3).

Lab Group: CSP (Labs 8–9)

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Lab 8 — Solving CSPs by Enforcing Arc Consistency

Goal: Solve constraint satisfaction problems via constraint propagation. **Steps:** Model a CSP (variables, domains, constraints); implement AC-3 arc consistency; pair with backtracking search (MAC); test on map coloring, N-Queens, Sudoku. **Deliverable:** AC-3 + backtracking solver + trace showing arc eliminations. **Resources:** AI/Lab-Resources.md → Lab 8

Lab 9 — Solving CSPs Using Stochastic Local Search Techniques

Goal: Solve CSPs with stochastic local search. **Steps:** Implement min-conflicts heuristic (N-Queens); compare with hill climbing, simulated annealing; observe scaling to large N (10K+ queens). **Deliverable:** Min-conflicts solver + performance comparison vs backtracking. **Resources:** AI/Lab-Resources.md → Lab 9

Exam Relevance (CSP)

- AC-3 (arc consistency) and CSP formulation = exam-relevant. See Formula-Book.md, Fear-Killer-Packs.md (AI-W4).

Lab Group: Expert Systems (Labs 10–12)

Source: Practical Workbook CS-323 (NED UET, revised Aug 2024). CLO: "Demonstrate the use of modern tools and techniques for developing intelligent systems. (C3, PLO-3)"

Lab 10 — Developing Knowledge-Based Systems

Goal: Build a knowledge-based (expert) system. **Steps:** Define knowledge base + rules; use an expert-system shell (CLIPS via clipspy, or PyKE); implement forward/backward chaining; classic animal-classification example. **Deliverable:** Working KBS + rule base documentation. **Resources:** AI/Lab-Resources.md → Lab 10

Lab 11 — Constructing Complex Rule-Based Systems

Goal: Scale to complex rule bases with conflict resolution. **Steps:** Large rule sets; conflict resolution strategies (refraction, recency, specificity); rule engines with RETE (Drools, CLIPS); rule matching efficiency. **Deliverable:** Complex rule engine + conflict-resolution example. **Resources:** AI/Lab-Resources.md → Lab 11

Lab 12 — Practicing Data-Driven Programming in Expert Systems

Goal: Data-driven (forward-chaining) inference in expert systems. **Steps:** Forward-chaining inference engine from scratch or via RETE library; diagnosis expert systems (medical/network/car); working-memory updates per firing. **Deliverable:** Data-driven inference engine + trace of rule firings. **Resources:** `AI/Lab-Resources.md` → Lab 12

Exam Relevance (Expert Systems)

- Forward vs backward chaining, conflict resolution, RETE = exam-relevant. See `Fear-Killer-Packs.md` (AI-W9).

Lab Group: Fuzzy Logic (Lab 13)

Source: Practical Workbook CS-323 (NED UET, revised Aug 2024). CLO: "Demonstrate the use of modern tools and techniques for developing intelligent systems. (C3, PLO-3)"

Lab 13 — Developing Fuzzy Logic Based System

Goal: Build a fuzzy logic system. **Steps:** Define membership functions (triangular, trapezoidal); write fuzzy rules; implement Mamdani inference + defuzzification (centroid); build a controller (temperature / washing machine / tipping) using scikit-fuzzy or MATLAB Fuzzy Logic Toolbox / Fuzzy Tech. **Deliverable:** Fuzzy inference system + input→output surface. **Resources:** `AI/Lab-Resources.md` → Lab 13

Exam Relevance (Fuzzy Logic)

- Mamdani vs Sugeno, defuzzification = exam-relevant. See `Fear-Killer-Packs.md` (AI-W12).

Viva — Artificial Intelligence (CS-323)

Project Walkthrough (3 min)

1. Which AI algorithm did you implement?
2. Show me your code running
3. What heuristic did you use? Is it admissible?
4. How does your GA ensure convergence?
5. What is the fitness function?

Technical Questions

6. Why A* instead of BFS for your problem?
7. Is your heuristic admissible? Prove it.
8. What happens if you change the crossover rate?
9. Explain backpropagation step by step
10. Why can't a perceptron learn XOR?
11. What is the difference between Mamdani and Sugeno?
12. How does forward chaining differ from backward chaining?
13. What is the significance of the learning rate?
14. Your neural network is overfitting. How do you fix it?
15. What is the vanishing gradient problem?

(Viva questions verbatim from `Viva-Book.md` — 5 walkthrough + 10 technical. Viva is a separate W15 slot.)